BUI THI QUYNH NHU

Data Engineering · AI Engineering · Business Analytics

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Github Profolio

SUMMARY OF QUALIFICATIONS

Computer Science undergraduate at Ton Duc Thang University with 7 end-to-end projects spanning deep learning, cloud infrastructure, real-time computer vision, and full-stack engineering. Demonstrated ability to train and deploy CNN/LSTM/YOLO models, architect containerised multi-service systems, and optimise data pipelines for performance. Seeking a Data Engineering or AI Engineering internship to drive analytical and predictive solutions at scale.

EDUCATION

Ton Duc Thang University — Ho Chi Minh City *Sep 2023 – Present*

Bachelor of Science in Computer Science · GPA: **3.2 / 4**

Relevant coursework: Machine Learning, Digital Image Processing, Database Systems, Data Structures & Algorithms, Cloud Computing, Operating Systems

TECHNICAL SKILLS

Languages & Query: Python (PyTorch · TensorFlow · Pandas · NumPy · OpenCV), SQL, JavaScript, PHP

ML / AI: CNN · LSTM · YOLO · Transfer Learning · Feature Engineering · Model Evaluation · Data Augmentation

Data & Cloud: Docker · AWS · Nginx · Linux · SQL Databases · REST APIs · DevOps · Cloud Security · Postman

Web & Tools: Node.js · HTML/CSS · Git · Real-time Monitoring · System Administration

Soft Skills: Analytical Thinking · Technical Communication · Project Management · Cross-functional Teamwork

Buiness Analysis: Requirement Gathering · Stakeholder Analysis · Use Case Modeling · UML (Sequence, Class) · ERD Design · Business Process Modeling · Agile Scrum · Test Case Design · System Analysis · Technical Documentation

RESEARCH & PROJECT EXPERIENCE

Visual Question Answering System *Feb 2026 – Mar 2026*

Deep Learning Research Project · Role: Algorithm Designer

- Designed a 3-component VQA architecture: **ResNet50** (visual feature extractor) + **BiLSTM** (bidirectional question encoder) + **LSTM decoder** (answer generation), achieving multi-class prediction across 3 answer categories.
- Leveraged pretrained ResNet50 ImageNet weights to reduce training time by **~40%** vs. training from scratch, with full benchmark comparison documented.
- Conducted **5-configuration ablation study** comparing BiLSTM vs. LSTM encoders, attention fusion strategies, and dropout rates — identifying optimal architecture.

Fruit Quality & Ripeness Recognition *Nov 2024 – Feb 2025*

Undergraduate AI Research · Role: Algorithm Designer & Researcher

- Built CNN-based classification pipeline detecting ripeness across **4 quality grades**, advancing to university competition finals — placing in the **top 15% of submissions**.
- Designed custom augmentation pipeline (rotation, colour jitter, crop) that improved out-of-sample generalisation by **~20%** on held-out test data.
- Reduced misclassification false-positive rate by **18%** through class-weighted loss functions and iterative hyperparameter tuning.

Minicloud — Cloud Infrastructure Simulator *Oct 2025 – Nov 2025*

Systems Engineering · Role: Full-Stack & Cloud Architect

- Architected and deployed a **9-service containerised cloud system** (Docker) replicating AWS compute, storage, and identity management.
- Engineered Nginx intranet routing layer sustaining **zero packet-loss across 100+ simulated concurrent connections** during stress tests.
- Implemented RBAC identity controls and real-time health dashboards, reducing mean incident response time to **under 2 minutes**.

Traffic Sign Recognition System *Oct 2025 – Nov 2025*

Computer Vision / Real-Time AI · Role: Image Processing Lead

- Built real-time video inference pipeline (OpenCV + YOLO) achieving **91%+ classification accuracy** across 4 sign categories at 24 fps live stream.
- Extracted **6 colour and 4 geometric features per frame**; integrated object tracking to maintain sign traces across frames.
- Reduced false detections by **25%** via per-class confidence threshold calibration using precision-recall curve analysis.

Computer Shop E- Commerce Platform *Aug 2025 – Sep 2025*

Full-Stack Web Development · Role: Full-Stack Engineer

- Developed e-commerce platform supporting **500+ SKUs** with dynamic filtering, multi-category management, and end-to-end order processing.
- Optimised SQL query execution by **35%** through index design and query restructuring on high-frequency product search endpoints.
- Implemented paginated REST APIs + caching layer, cutting average page load from **1.8s → 0.7s**.

TechSync Store — BA Case Study *Aug 2025 – Sep 2025*

Business Analyst

- Conducted user survey and stakeholder analysis to define system requirements
- Designed UML diagrams (Use Case, Sequence, Class) and ERD
- Defined functional & non-functional requirements
- Built test plans and system-level test cases
- Mapped user flow: Browse → Cart → Checkout →Order Management
- Agile Scrum with sprint planning & requirement traceability matrix

Real-Time Note-Taking Web Application *Apr 2025 – May 2025*

Full-Stack / Cloud Deployment · Role: Full-Stack Engineer

- Deployed cloud-hosted application achieving **99.5% uptime** over a 3-week live testing period with concurrent read/write operations.
- Designed RESTful API backend with optimistic locking, eliminating data conflicts under concurrent access.

PROFESSIONAL EXPERIENCE

Mathematics, Science & English Tutor *Nov 2024 – Jun 2025*
eTeacher Company · Ho Chi Minh City

- Delivered **in-home Mathematics and Science tutoring entirely in English**, strengthening both subject fluency and English communication skills for high school students.
- Conducted **online English language classes** via digital platforms, adapting lesson delivery for remote learners across varying proficiency levels.
- Designed structured, step-by-step problem-solving frameworks — analytical skills directly applicable to data pipeline debugging and root-cause analysis.

CERTIFICATIONS & ACTIVITIES

IELTS 6.5 · Aptis 163 (Advanced English) *2024 – 2025*
Member — Technology Club (Programming · AI · Robotics) *2023 – 2025*